

popquiz power, conservation of energy , food Cal

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:41pm

Quiz Instructions

ONE ATTEMPT!!!



Question 1 17 pts

Each year 500 runners run up the stairs to the 86th floor of the Empire State Building in New York City. There are 1576 steps and each step is 0.28 m high. In 2003, Australian Paul Crake (20-29 age group) set the overall record by reaching the 86th floor in 9:33. His mass was 70.0 kg.

Say the time was 9 minutes . What is the energy he produced during this even? in J

hint: $PE = mgh$ and you have the height of each stair. take $g=10\text{m/s/s}$

1K is 1000 and 1M is one million



about 1MJ



about 309KJ



about 31K



about 200J



Question 2 17 pts

Building on the previous question. What is the energy in Cal (food calories)

1 Cal is 4184J.

round to nearest ten



about 7000 Cal



about 70 Cal



about 240Cal



about 2000 Cal



Question 3 17 pts

Building on the previous question

what is the power produced by the racer?



about 1500W



about 30,000 W



about 200W

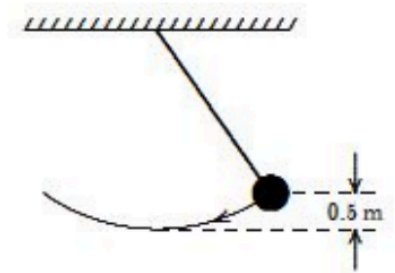


about 570W



Question 4 17 pts

36. A simple pendulum consists of a 2.0-kg mass attached to a string. It is released from rest at X as shown. Its speed at the lowest point Y is about:

☐ (A) 0.90 m/s☐ (B) 1.90 m/s☐ (C) 3.1 m/s☐ (D) 6.0 m/s☐ (E) 36 m/s

A



E



B



D



C



Question 5 17 pts

In your body you have 7 Cal per gram.
So if you cut your Calories from 2,000 to 1,500 calories per day.

How many pounds do you lose per week ? (**take 450g = 1 pound about**)

hint: first find the grams. then the pounds

we supposed you have a good metabolism (no insulin resistance). There sre 7 days in a week

reference: Physics behind the headlines, Richard Muller

☐

about 1 pound

☐

about 5 pounds

☐

about 0.5 pounds

☐

about 10pounds



Question 6 17 pts

in one half-hour of exercising , you lose 150 Cal about. (same Cal in a soda/orange juice).

If you exercise one hour everyday, how many pounds you lose per week?

hint: use 7Cal in 1g.

(suppose a good metabolism. its an approximation)

☐

about 1kg

☐

about 300g

☐

about 500g

☐

about 700g

Quiz saved at 7:41pm

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